25.9.2026

Practical-9

Implementation of subnetting

AIM:

Implementation of subnetting in CISCO PACKET TRACER simulator.

INTRODUCTION:

subnetting is the process of dividing a large IP network into smaller logical networks called subnets. classless subnetting allows us to use subnet masks other than default class-based masks, making IP address more efficient and allowing multiple networks to be created.

Here the nw 192.168.1.0/24 is divided into smaller /27 subnets. A /27 subnet provides 30 usable host address per subnet.

PROCEDURE:

1) open cisco packet tracer
2) Add required routers, switches and PCs
3) connect the devices using appropriate cables.
4) configure the routers with required IP address and subnet masks.
5) configure the router with CLI
   - enable
   - configure Terminal
   - select required Fast Ethernet / Gigabit Ethernet interface
   - Assign IP address and subnet mask
   - use no shutdown to activate interface.

6) Configure switch ports in access mode wher. required.

7) Assign static IP address, subnet mask and gateways

8) Use /27 subnet mask: 255.255.255.224

9) Configure four subnets:

192.168.1.0/27

192.168.2.0/27

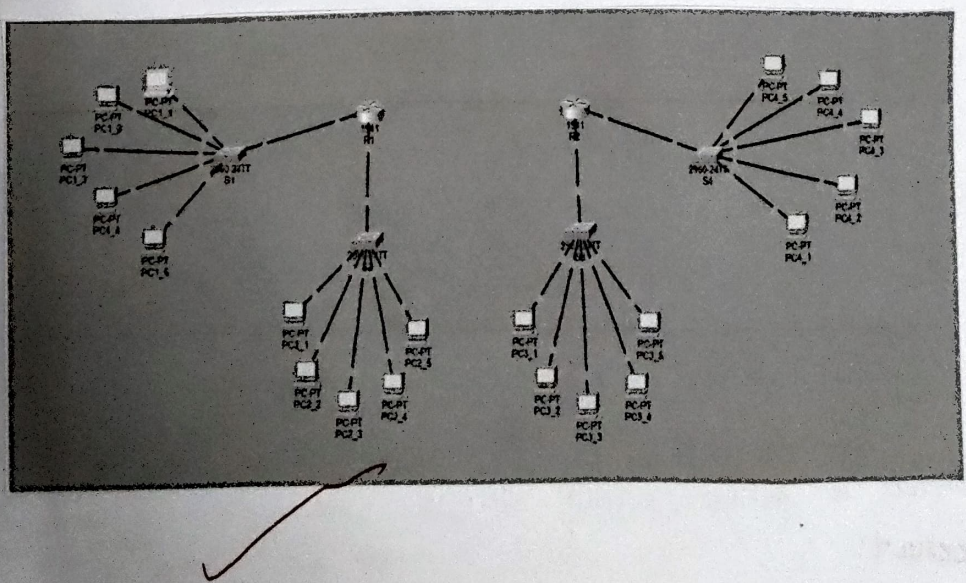
192.168.3.0/27

192.168.4.0/27

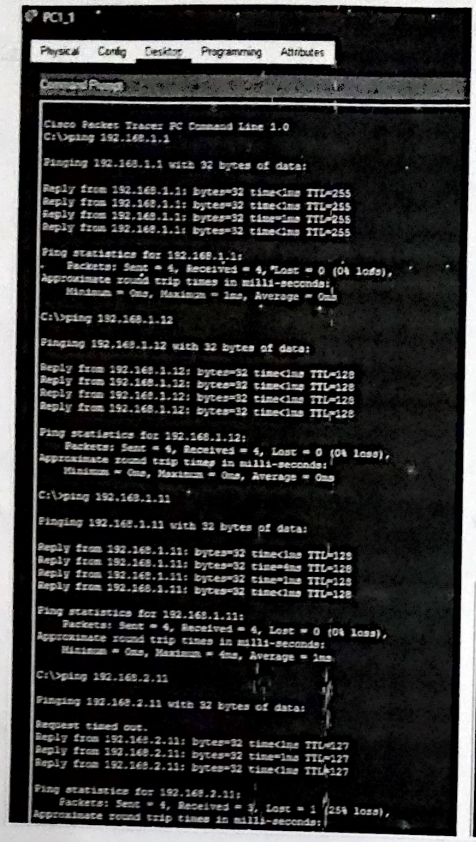
10) Verify router interface and PCs belong to respective router.

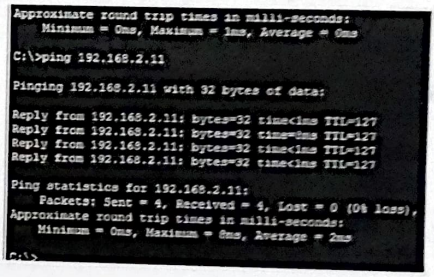
11) Open command prompt to PCs

12) Using ping command to test connectivity between PC and respective router.



Ping used to verify communication between PCs in different subnets.





RESULT:

Thus subnetting was successfully implemented using cisco packet tracer.